Project and Script Guide

A practical explanation of the final FIREWATCH SA application

STAT7006A | Spatial Statistics Dashboard | Final bright-theme candidate

1. Project at a glance

FIREWATCH SA — Where South Africa Burns is a seven-tab R Shiny dashboard developed for a spatial statistics group assignment. It uses the group's existing 2024–2025 VIIRS NOAA-21 active-fire detection dataset and South African provincial polygons. The interface uses a bright ivory, orange, coral, gold and teal theme. The app can explore one province or the country as a whole.

Element	Final implementation
Research focus	Where and when satellite active-fire detections occur, and how their point patterns compare with uniform spatial randomness.
Dataset	NASA FIRMS VIIRS NOAA-21, 2024–2025; existing prepared sf points and province polygons.
Methods	Kernel Density Estimation; Nearest Neighbour with CSR Monte Carlo test; Ripley's K displayed as L(r)−r.
Geographic scope	One of nine provinces or All Provinces (national).
Hosting	https://brudance.shinyapps.io/firewatch-sa/
Project status	Final candidate reported working locally and online by the team; runtime RDS files must be supplied to run from this ZIP.

2. The seven tabs

Tab	What a user can do	What to explain
Start Here	Read context, purpose and navigation.	Fire detections are a starting point for spatial investigation, not a direct measure of fire damage.
Overview	Filter province/year/season; view map, headline counts, charts; download CSV.	Distribution and totals change with filters. The point map is visually sampled for performance.
Seasonal Fire Intelligence	Compare monthly/seasonal 2024 vs 2025 counts and province–season heatmap; download summaries.	This tab compares both years deliberately, so it does not obey the sidebar Year and Season selections.
Kernel Density	Choose bandwidth (2–30 km), run KDE, download PNG.	Smoothed concentration/intensity of detections, not a test of statistical significance.
Nearest Neighbour	Run observed/expected distance, ratio and 99 CSR simulations; export CSV.	NN ratio below one indicates shorter average distances than simple CSR expectation; p-value tests clustering.
Ripley's K	Run K with 19 CSR simulation envelopes; download curve CSV.	L(r)−r describes clustering or dispersion at multiple distance scales.
Methods & Limitations	Read assumptions, projections and caveats.	Explain why satellite detections and uniform CSR require careful interpretation.

3. How the R script is organised

Packages and startup checks

The script loads shiny, sf, dplyr, ggplot2, leaflet, scales and spatstat components. It checks for data/fires_final.rds and data/provinces.rds and reads them as sf objects. Missing files cause startup to stop clearly.

Data preparation at application startup

Fire and province geometries are transformed to EPSG:4326 for mapping. Longitude/latitude are extracted once. Monthly and seasonal summaries are pre-aggregated at startup so repeated chart requests avoid costly spatial operations.

Spatial-analysis helper

make_point_pattern() samples points reproducibly (seed 123), projects them into metres, constructs a study window from the selected provincial/national polygons and creates a spatstat ppp object. It records the number of points analysed and any duplicated coordinates.

Interface (ui)

A fixed sidebar controls province, year and season. The tabsetPanel contains seven named tabs. CSS sets the bright visual theme; shiny outputs are placeholders that update when server functions run.

Server reactives

filtered_fires() follows the sidebar selections. selected_boundary() selects either a province polygon or all provinces. Analysis results are computed only when the user clicks each method's Run button. filters_match() prevents stale outputs being mistaken for results from changed filters.

Downloads

The Overview and Seasonal Intelligence tabs export CSV summaries; KDE exports a PNG; NN and Ripley's K export results as CSV. These are generated from the current calculations.

Publishing

The R script ends with shinyApp(ui = ui, server = server). Deployment is done separately from the RStudio Console, not from within app.R.

4. Spatial statistics: calculations and interpretation

Kernel Density Estimation (KDE)

The app estimates a smooth detection-intensity surface using spatstat.explore::density.ppp(). The user controls the Gaussian bandwidth from 2 to 30 km. It uses at most 6,000 reproducibly sampled points and rescales the sampled density to the full filtered detection count. Output is displayed as a coloured surface and can be downloaded as a PNG.

Interpretation: brighter/higher-intensity areas have more satellite detections concentrated nearby at the chosen smoothing scale. A wider bandwidth produces broader, smoother patterns; a smaller bandwidth shows more local detail. KDE is descriptive, not a formal hypothesis test, a burned-area estimate or a fire-severity map.

Nearest Neighbour (NN)

The app calculates the mean observed nearest-neighbour distance and compares it with an infinite-plane uniform-CSR approximation: $E(d) \approx 1 / (2\sqrt{\lambda})$, where λ is sampled point count divided by study-window area. The nearest-neighbour ratio is observed / expected. It then simulates 99 uniform random point patterns with the same number of points in the same study window. The one-sided Monte Carlo p-value is (1 + number of simulated mean distances ≤ observed) / 100.

Interpretation: ratio < 1 suggests clustering relative to the benchmark; ratio ≈ 1 is near the benchmark; ratio > 1 suggests more dispersion. A small one-sided p-value provides evidence of clustering relative to the simplified uniform-CSR model. The test does not establish causes or prove clustering of independent wildfire events. The planar expected-distance approximation does not fully adjust for edge effects.

Ripley's K (transformed to L(r)−r)

The app estimates border-corrected K(r) for distances from 0 to 10 km in 0.5-km increments, then displays $L(r)-r = \sqrt{(K(r)/\pi)}-r$. It compares the observed curve with pointwise envelopes from 19 uniform-CSR simulations, using at most 800 sampled detections.

Interpretation: positive L(r)−r indicates more nearby pairs than the uniform-CSR reference at that distance; negative values indicate fewer. A curve above the simulated pointwise band is an exploratory indication of excess clustering at that distance. These are pointwise envelopes—not a global p-value or a formal overall significance test.

5. National versus provincial analysis

When All Provinces is selected, the script dissolves provincial boundaries into a national study window and projects data to a South Africa-centred Lambert azimuthal equal-area (LAEA) system. Individual provinces use an appropriate southern-hemisphere UTM zone. Distances are calculated in metres. National distance distortion is not zero, and national and provincial analyses should not be compared without considering the different window sizes and sampled point counts.

Method	Max sampled points	Reference calculation
KDE	6,000	Gaussian kernel; edge correction; bandwidth 2–30 km; density rescaled to filtered count.
NN	2,000	Observed/expected mean distance; 99 CSR simulations; one-sided clustering p-value.
Ripley's K	800	Border-corrected K; 19 pointwise CSR envelopes; distances 0–10 km.

6. Seasonal intelligence: reading the charts

The dashboard precomputes month-level counts for 2024 and 2025 and displays monthly lines, seasonal bars and a province-by-season heatmap. The seasonal tab respects the Province filter, but deliberately ignores the Year and Season filters so it can compare the complete calendar years. Summer is December–February, Autumn March–May, Winter June–August and Spring September–November; December is assigned to its calendar year, not to a cross-year fire season.

A difference in the number of satellite detections is not necessarily a change in the number of distinct fires or burned area. Satellite overpass timing, observation opportunity, cloud/smoke conditions and landscape differences can affect detections.

7. Technical limitations and responsible claims

- Each record is an active-fire detection (thermal anomaly), not a unique fire incident; one event can generate multiple detections.
- Unadjusted counts are not normalized by province area, available burnable vegetation, or satellite observation opportunity.
- The simple uniform complete spatial randomness null ignores environmental heterogeneity and repeated observations.
- Reproducible random sampling improves speed but means analyses are performed on subsets, not every record.
- KDE intensity surfaces are not causal maps; NN clustering p-values and pointwise K envelopes should not be over-interpreted.
- National analysis uses an equal-area projection, but distance distortion and scale effects remain.
- Live hosting may time out when idle; refresh the dashboard and allow it to reconnect.

8. Code handover and verification

1. Copy app.R and both RDS files into the same project structure (see README).
2. Install the packages listed in README and run app.R locally.
3. Check the Start Here and Seasonal Fire Intelligence tabs.
4. Test KDE, NN and Ripley's K for a province and (if time permits) All Provinces.
5. Change a filter after a run: the dashboard should ask you to rerun rather than display stale results.
6. Confirm the data and export files have the expected names and contents.
7. Publish only after local testing; run rsconnect::deployApp() in the Console, never in app.R.

Final selected version (23 October 2026): The KDE panel now displays an interactive OpenStreetMap basemap beneath the density raster. A separate static PNG download is retained. See Document 04 for implementation details, installation, testing, and caveats. This addition supersedes earlier descriptions of a KDE plot without a basemap.